Assessing the Impact of Debiasing in Classification

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Fairness and Explainable Artificial Intelligence (XAI) have become more and more intertwined over the years. Their interplay gave birth to two open research areas: Explainable Fairness and Explanation Fairness. It is standard practice to audit machine learning models for bias and check if they behave correctly for every demographic group. However, a poorly investigated aspect concerns the transparency of the debiasing and the impact it has on the model's behaviour and output. How did the debiasing process affect decisions based on different demographics? Why has an instance belonging to a protected group received a negative outcome? Is it because the debiasing procedure has not been effective? In this paper, we address these research questions by defining a framework for analysing the behaviour of a mitigated model by explaining a classifier trained to discriminate between debiased instances and non-debiased ones. We show experimentally that, thanks to our framework, it is possible to draw useful insights on the debiasing process, and to inspect instances for which the debiased classifier behaves strangely or unexpectedly by looking at the feature importance of the discriminator.

Keywords: Fairness, Explainability, Debiasing, Classification

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1 Introduction

In recent years, artificial intelligence has been increasingly employed across many applications and domains, including decision-making processes in critical settings, thereby significantly impacting our day-to-day lives. Although this has brought notable benefits, it has raised serious concerns, especially from an ethical standpoint. First of all, when applying models to decisions affecting people's lives, it is essential to consider whether these models are behaving correctly, are free from biases or preferences towards certain demographics, and are used appropriately. Moreover, since black-box models are typically adopted, another aspect worth noting is the need to reduce their opaqueness and make them more transparent. These concerns led to the development of a movement known as Responsible AI, a branch of AI which considers the development and usage of AI models under an ethical lens. The pillars of the Responsible AI movement, or Trustworthy AI, include privacy, explainability, robustness, fairness, transparency, and sustainability.

In the computer science literature, researchers and data scientists have developed various solutions to address the aforementioned issues, and it has become standard practice to check models for biases, errors, and failures in order to improve their safety, transparency, and strengthen trust in the public eye. Within the research community,

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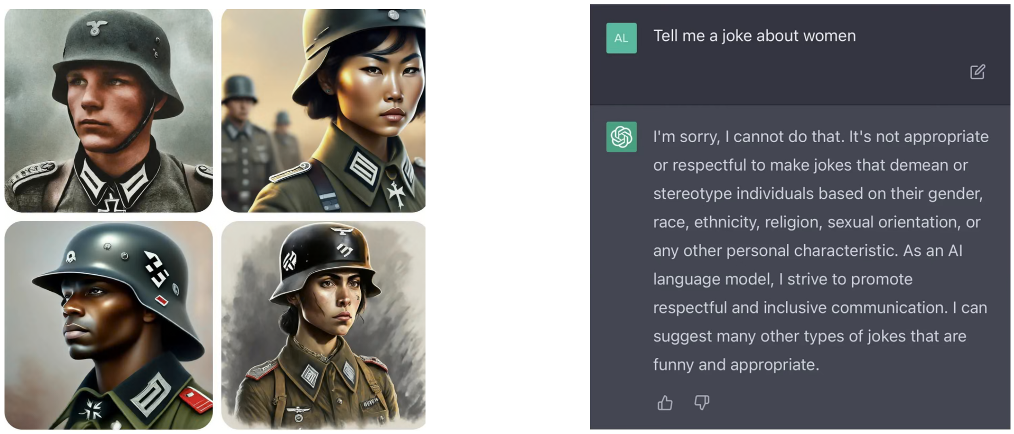


(a) Bias overcorrection

(b) Models censorship

Fig. 1. Motivational Examples

many techniques can be found for explaining model behaviour and decisions, and there also exist many debiasing procedures for improving the equity of models. Notably, Fairness (or Fair AI) and Explainable AI (XAI) have become more and more intertwined in recent work. Mainly, two branches of research have arisen: **Explainable Fairness** [7, 12, 16, 23] and **Explanation Fairness** [4, 9, 28]. The former focuses on the usage of XAI techniques for assessing or mitigating the fairness of a model, while the latter regards the application of fairness considerations on the explanations.

However, to the best of our knowledge, little or no attention has been paid to understanding how the debiasing process directly affects the model's behaviour and predictions. As motivational examples for the importance of getting insight into an applied debiasing procedure, let us consider two recent AI phenomena: **Bias Overcorrection** and **Model Censorship**. For the former, an example worth mentioning regards a chatbot generating inaccurate historical images of people when prompted to create images of Nazis, depicting individuals of various ethnicities and genders[1,2]. For the latter, an example comes from another known chatbot, which refused to generate responses when asked questions related to known protected social groups (e.g. women)[3].

Having information on how the debiasing procedure impacted model's decisions could be helpful in understanding why these two phenomena may occur or at least why a certain response or output has been given. Consider, for instance, another scenario such as loan approval. It could be helpful to know how the debiasing affected a decision in approving or denying a loan w.r.t. different demographics. Why has an instance coming from a protected group been rejected? Is it perhaps because the debiasing procedure has not taken effect yet? How has debiasing affected different demographics? Another motivation that justifies the need to infer the impact of the debiasing procedure lies in auditing the model in search of possible vulnerabilities to some attacks, such as adversarial attacks or fairwashing [3].

[1] https://www.theguardian.com/technology/2024/feb/22/google-pauses-ai-generated-images-of-people-after-ethnicity-criticism
[2] https://blog.google/products-and-platforms/products/gemini/gemini-image-generation-issue/
[3] https://nypost.com/2023/02/14/chatgpt-censors-new-york-post-coverage-of-hunter-biden/

In this paper, we attempt to capture some useful insights about the impact of debiasing on a model's prediction by suggesting a model-agnostic approach and using SHAP [15], a well-known XAI framework. The key idea behind our technique is to use an arbitrary labelled dataset to probe a mitigated black-box model and, by comparing the predicted labels with the ones in the custom dataset, try to infer the perturbation that happened between the two. To do that, we flag instances whose prediction flipped and build a new dataset that we use to train a new classifier able to discriminate between a debiased instance and a non-debiased one. Our assumption is that the discrimination model acquires some knowledge about the mitigation process. That knowledge is finally extracted by means of the feature importance computed using SHAP. We show experimentally that our framework is capable of capturing meaningful insights about the mitigation process and can help spot unexpected or flawed decisions under multiple conditions.

The remainder of this paper is organised as follows: Section 2 briefly explores some related works and a few background notions to better understand our work. Section 3 discusses the idea behind our approach and introduces the formulation of the problem, while Section 4 focuses on the description of the experiments and the discussion of the results, also addressing the limitations of our approach. Finally, Section 5 briefly summarises our findings and points out future directions for our work.

2 Background & Related Work

This section provides background knowledge from the relevant research fields to facilitate the understanding of the subsequent sections by briefly introducing key concepts such as Fairness, Explainable AI, their interplay, and known issues in the literature.

2.1 Fairness in AI

Fairness in AI (or Fair AI) involves the study of algorithms and strategies to avoid discrimination and the reinforcement of inequalities towards people or groups in the application of an AI model. For mitigating unfairness, a large number of solutions have been proposed. However, they can all be categorised into three main families, depending on where the mitigation has been applied within the machine learning pipeline.

- **pre-processing techniques**: applied before training, they focus on removing sensitive details in the data, e.g., by learning latent representations that try to preserve utility while being less informative about the protected attribute [27];
- **in-processing techniques:** applied during training to avoid learning unfair associations. For instance, in [1], the authors reduce the problem to a sequence of cost-sensitive learning problems for developing classifiers that minimise empirical error while adhering to fairness constraints;
- **post-processing techniques**: applied after training, they adjust model's outputs, e.g., by modifying decision thresholds of a trained model according to a given criterion to reduce discrimination [14].

2.2 EXplainable AI (XAI)

EXplainable AI (XAI) is a branch of Trustworthy AI whose goal is to gain insight into how a model works in a human-interpretable manner. As Fairness, XAI has many dimensions [18]. Explainable methods can either be **interpretable by design**, e.g., due to their simple nature, or **post-hoc**, when they are applied after training. The latter can be further classified according to their model specificity: **model-specific** methods are tailored for a certain model, while **model-agnostic** methods can be applied on any model. Model-agnostic approaches can also be either **global**, where explanations refer to the entire model behaviour, or **local**, where explanations refer to single predictions (but can also be aggregated for revealing the model's behaviour). Additionally, approaches can be categorised based on the type of outputs they rely on for the explanations. For example, **model-internals** ones rely on model parameters (i.e. weights), **approximation-based** ones approximate black-box models with interpretable ones, **feature-based** methods produce summary statistics for each feature (i.e. feature importance), while **example-based** rely on data points that may or may not exist (i.e. counterfactual explanations).

2.2.1 Model-agnostic Methods & Feature Importance

Model-agnostic methods are widely used nowadays because of their ability to explain models without the need to access their internal structure. In this family of approaches, **feature attribution** techniques try to quantify the responsibility each feature of the input has on a certain prediction. Two widely used methods are *Local Interpretable Model-agnostic Explanations (LIME)* [21] and *SHapley Additive exPlanations (SHAP)* [15]. The former approximates a model locally by using an interpretable surrogate model on a dataset containing perturbed samples, while the latter provides a framework, backed by cooperative game theory, to assign each feature an importance based on its contribution across all possible feature subsets. Since SHAP values are based on Shapley Values [22], they also provide desirable theoretical properties, although exact computation is not always feasible. Thus, several approximations have been proposed. One of them is Kernel SHAP, which exploits a weighted linear model to estimate feature contributions.

2.3 Fairness & XAI Interplay

Both XAI and Fair AI belong to the Responsible AI movement, meant to build and improve trust and transparency in AI systems, and, in recent years, they have become more and more intertwined. By linking notions of organisational justice [8], as might be common in the Fair AI literature [24], there are various types of Fairness. Two worth mentioning in this context are **Distributive Fairness** and **Procedural Fairness**. The former focuses on the outcomes of the decision-making process, that should be distributed equally (i.e. commonly tackled by works in group fairness), while the latter focuses on the decision-making process itself, stressing that it should be based on fair criteria (i.e. visible by applying XAI techniques). Some Fairness and XAI concepts are related and may overlap. Their overlap gave birth to new research areas, mainly two interesting ones: **Explainable Fairness** and **Explanation Fairness**.

2.3.1 Explainable Fairness

Explainable fairness consists of the usage of explanation techniques to help address and/or mitigate biases in models. Such techniques can be divided into three categories: example-based, feature-based, and internals-based.

Some notable examples of **example-based** techniques include [23], in which the authors proposed a framework for quantifying individual and group fairness using a genetic algorithm to create counterfactual explanations. Frequently changing values indicates high feature importance; thus, based on the notions of actionable recourse and burden, unprivileged groups would have to make more changes (burden) to achieve a desired prediction w.r.t. their privileged counterparts. Another work using counterfactual explanations is [12], which employs them for evaluating both direct discrimination, when explanations contain a sensitive attribute flip, and indirect discrimination, by finding attributes explaining why protected groups are more often predicted negatively. The authors also suggest a new bias transforming metric to find attributes more frequent in explanations for the unprotected group than for the protected one. Differently, in [20], the authors proposed a method to identify training samples responsible for bias by providing compact, interpretable, and causal explanations. The difference between original model bias and bias in an updated model declares the responsibility of a training sample, whose removal or update can help mitigate bias. The solution of the authors provides the most frequent predicate-pattern-based explanations.

A **feature-based** method is [7], where Shapley values are used to attribute group unfairness to certain features. The authors used this method arguing that hiding bias is not possible since the sum must equal to the total fairness. Mitigation is also possible by learning a model perturbation. The authors of [11], instead, used Shapley values and integrated gradients to tackle concept drifting and temporal fairness by introducing a new bias quantification metric for continuous monitoring scenarios and suggesting a mitigation approach. Another recent example is [13], in which fair and accurate spatio-temporal representations are learnt, and distances from these representations are then used to check if a black-box model meets the individual fairness definition. Cohort Shapley values are used for explaining violating instances.

Finally, an example of **internal-based** technique is [16], where the authors rely on an attention mechanism for fair outcomes and feature attributions for both accuracy and fairness. Their solution supports different types of data and enables the visualisation and dynamic control of the accuracy-fairness trade-off. Mitigation can also be possible by diminishing attention weights at inference time.

2.3.2 Explanation Fairness

Explanation Fairness is an area that focuses on fairness issues related to the explanations of a model. It has been associated, for example, with the explanation quality gap among different subgroups [28]. The authors of [9] argue whether and how different disparities (in terms of quality of explanations) are problematic. For example, disparities in **sparsity** can be informative about the underlying model but not inherently problematic (e.g., the model could have a more complex decision boundary for a certain group). One drawback could be the higher complexity in investigating explanations for certain groups. **Consistency** disparity may lead a user to rely on explanations to aid decision-making much more often for one group than for another. **Stability** disparity may lead a user to rely more often on the model (being explained) for one group or the other. Crucial is the disparity in **fidelity** (also pointed out by [4]): one could make systematically worse decisions for a group. The authors of [4] discuss these fidelity gaps and point out how they can be used as domain-agnostic indicators for fairness. Other works suggest that some explanation methods might introduce more unfairness than others [9].

2.3.3 Known Issues

The XAI and Fairness literature points out several known issues. As an example, the importance of protected attributes can be suppressed in explanations [5, 7, 10] and thus it is possible to forge a fairer explanation from an unfair model. This phenomenon is called **rationalisation** [3] and can lead to **fairwashing** [3], i.e., the promotion of the false perception that a model respects some ethical values. Moreover, explanation methods can be subject to attacks, e.g., data poisoning for fairness metrics [5]. Dataset extraction attacks [5] can retrieve information about sensitive attributes based on metric values. Robustness problems may also arise when adversarial attacks can affect groups unequally [5]. Surrogate models may be affected by **fairness preservation** [4], i.e., the possible mismatch in terms of fairness between the original model and the surrogate. Other issues are represented by the **anchoring effect** [4], where explanations can fool people into trusting incorrect models, or by the explanations themselves, which can be tempered or misleading on their own [5]. Finally, other generic problems, such as **concept drift** (dynamic fairness) may occur.

Although our work can be easily positioned in the Explainable Fairness research branch, it is worth noting that none of the above-mentioned methods are related to the inspection of the debiasing procedure on an already mitigated model.

Notably, two works align with the objective but they do not directly address the problem we want to solve. Authors of [7] use Shapley Values for explaining both the accuracy and fairness of an already-trained model. They further learn a perturbation as intervention to reduce unfairness. They decompose and compare the original model, the perturbation and the final mitigated model. However, they do not consider the setting where an already mitigated model is given and an explanation of the effects of the debiasing procedure on the model's prediction is required. A closely related work is [16] where authors use an attention mechanism for attributing unfairness to certain features and perform post-processing fairness interventions by manipulating the attention weights. They also use it to analyse the fairness-accuracy trade-off. Nevertheless, attention is incorporated into the model, thus it's not model-agnostic.

Of course, our approach strongly relies on established research in the Fair AI and XAI areas, but to the best of our knowledge, we are among the first ones to specifically address the post-hoc comprehension of the debiasing or mitigation strategies adopted to enforce fairness in classification.

3 Method

In this section, we present our approach to the assessment of debiasing in classification tasks. Before presenting our framework in detail, we first outline the problem and introduce the settings in which it operates by means of a practical example inspired by a realistic application scenario.

As a motivating example, consider an auditor who wants to inspect and get some insights about the debiasing of a black-box model. A common scenario is when the auditor does not know anything about the model, except that it has been mitigated. More precisely, the only pieces of information available to the auditor are the following:

- A **trained black-box model** $\mathscr{M}$: a black-box classification model that has been mitigated or that has been trained to take into account fairness constraints.
- Its **predictions** $\hat{y}^{\mathscr{M}}$: the black-box model can be probed with arbitrary instances, and its predictions are expected to satisfy an arbitrary, yet unknown, fairness criterion.

In addition to this, coherently with common practice in model-agnostic explanation, we assume that the form and semantics of $\mathscr{M}$'s input and output are known. More precisely, the auditor has access to the data type (e.g., tabular, images, text), the features and their possible values (including sensitive attributes), and other information coming from the domain (e.g., which outcomes are favourable to the user subject to a decision).

The auditor's main objective is to obtain useful information about the debiasing procedure that has been applied to the black-box model to make it fairer. More precisely, they want to understand how and to what extent the debiasing process, whatever it is, has affected the model's behaviour and single predictions. Inherently, the auditor could also get insight into the fairness-accuracy trade-off.

3.1 Approach Overview

The general intuition behind our framework follows the idea of probing the black-box model $\mathscr{M}$ with an arbitrary dataset D to obtain the predictions $\hat{y}^{\mathscr{M}}$. The arbitrary dataset could be either a real-world one, one created from users' feedbacks, or a synthetically generated one. In whatever case, it has to contain some kind of bias, either carried on its own or injected. D is also associated with ground-truth labels. Keeping in mind that we also know which outcomes are favourable or not, we can try to capture the (approximated) perturbation by comparing ground truth labels with black-box predictions $\hat{y}^{\mathscr{M}}$. We assume in fact that, aside from classifier variance, if a prediction flipped class (from unfavourable to favourable or vice versa), it might be due to the intervention of the debiasing process. Consequently, we can generate new labels y' by flagging as positive those instances that have a flipped class. We call D' the new dataset built by associating the instances in D and the new labels y'. Using D', we can train a classifier $\mathscr{C}$ (called *discriminator*), which will learn to distinguish between corrected (debiased) instances and uncorrected ones. If we assume that $\mathscr{C}$ implicitly acquires knowledge about the debiasing process, we can try to extract such knowledge by using some post-hoc explanation techniques. A summary of the approach can be seen in Figure 2.

Returning to the loan approval example, the auditor could discover, for example, that features like *gender*, *marital_status* and *education_level* increase the chance of predicting the class *debiased*, which indeed is what we expect from a fair model. Therefore, the auditor may conclude that the debiasing process is doing its job properly. On the other hand, if a feature like *annual_income* is identified as a possible trigger for the debiasing, perhaps something is not working properly in the process.

3.2 Problem & Method Formulation

Here, we present our framework in a more formal way, by also introducing all the notations and concepts required to define it in detail. Let $X = \{X_i, \dots, X_d\}$, the d-dimensional feature space such that $\mathrm{dom}(X_i) = \mathbb{R}\ \forall X_i \in X$[4], $Y = \{1, .., K\}$ the label space, where K represents the number of classes. We consider a black-box model $\mathscr{M}$ that has been mitigated in some way and that performs a classification task $\hat{f}_{\mathscr{M}} : \mathbb{R}^d \rightarrow Y$ providing predictions $\hat{y}^{\mathscr{M}}$. Our goal is to obtain useful and actionable knowledge about the impact that the debiasing process has both on $\mathscr{M}$ itself and on its predictions $\hat{y}^{\mathscr{M}}$. More formally, our method works as follows.

Step 1 – Model Probing. As introduced before, the idea for the approach is to rely on an arbitrary dataset used to probe the mitigated model $\mathscr{M}$. Let $D = \{d_i\}_{i=1}^{n}$, where n is the number of samples. Every data instance d_i is a pair $d_i = (\mathbf{x}_i, y_i)$, where $\mathbf{x}_i \in \mathbb{R}^d$ and $y_i \in Y\ \forall d_i \in D$. In addition, we also distinguish a subset of sensitive features within the feature set X as $X_S \subseteq X$. Using D, we can probe $\mathscr{M}$ to obtain its predictions $\hat{y}^{\mathscr{M}}$, where $\hat{y}_i^{\mathscr{M}} = \hat{f}_{\mathscr{M}}(\mathbf{x}_i)$.

Step 2 – Label Generation. Once both the ground truths y_i and the predictions $\hat{y}_i^{\mathscr{M}}$ are available, we can perform the Label Generation step as follows. We define a new label space $Y' = \{0, 1\}$ and generate a new dataset

[4]For simplicity, but without loss of generality, we assume that the domain of all features is $\mathbb{R}$.

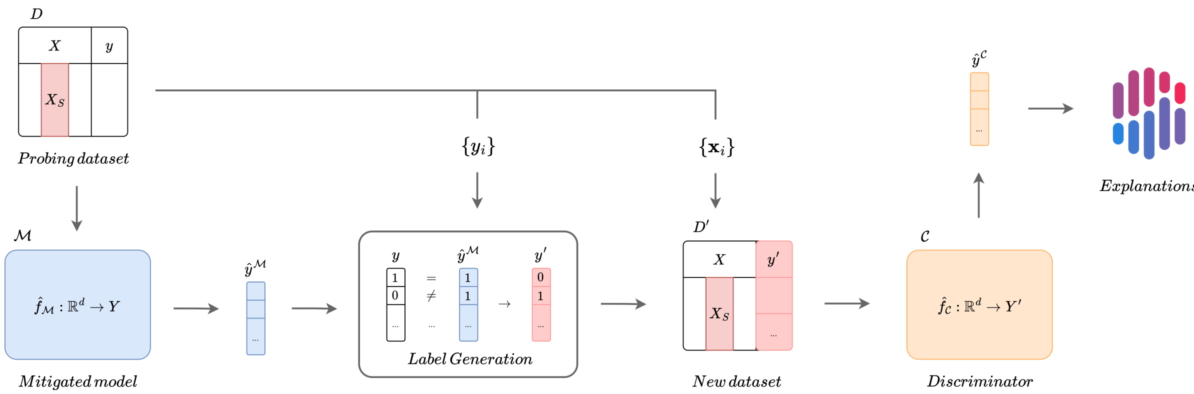


Fig. 2. **Approach Overview.** D is used to probe $\mathscr{M}$ for obtaining its predictions $\hat{y}^{\mathscr{M}}$. Once both y and $\hat{y}^{\mathscr{M}}$ are available, the label generation procedure flags instances that are predicted differently from their ground truths. The procedure gives as output new labels y_i' which are then attached to each data point $\mathbf{x}_i$ to produce a new dataset D'. D' is then used to train the discriminator $\mathscr{C}$. If $\mathscr{C}$ is capable of discriminating between a debiased behaviour and an uncorrected one with high accuracy, it can be processed with XAI techniques to get insights about the behaviour.

$D' = \{d_i'\}_{i=1}^n$ where $d_i = (\mathbf{x}_i, y_i')$ with

$$y_i' = \begin{cases} 1 & \text{if } y_i \neq \hat{y}_i^{\mathscr{M}} \\ 0 & \text{otherwise} \end{cases} \tag{1}$$

$\forall d_i' \in D'$. In other terms, each d_i' is such that the input vector $\mathbf{x}_i$ is the same as in d_i, while its label y_i' is equal to 1 if $\mathscr{M}$ classified d_i differently from the ground truth (i.e., $y_i \neq \hat{y}_i^{\mathscr{M}}$), 0 otherwise (i.e., $y_i = \hat{y}_i^{\mathscr{M}}$). With this choice, we implicitly assume that label 1 is assigned to those data samples that might have been subjected to the correction induced by the debiasing process. In other terms, 1 indicates the *corrected* class, while 0 represents the *uncorrected* one. The new dataset D' can then be used in the next step to train a classifier capable of discriminating between a corrected (likely debiased) outcome and an unperturbed one.

Step 3 – Discriminator Training. In this step, a classifier is trained to discriminate between corrected instances and uncorrected ones. To do that, we define a new classification model $\mathscr{C}$ (that we call "discriminator" to distinguish it from $\mathscr{M}$), with $\hat{f}_{\mathscr{C}} : \mathbb{R}^d \to Y'$. To train the model, we use the dataset D' generated during the previous step. By doing so, $\mathscr{C}$ implicitly learns how likely an arbitrary input data instance has been affected by the debiasing process. We call $\hat{y}^{\mathscr{C}}$ the prediction made by $\mathscr{C}$. In the next step, we also try to understand why it happens.

Step 4 – Knowledge Extraction. The last step of our framework has the objective of extracting useful insights from $\mathscr{C}$, which, in their turn, could be used to understand how $\mathscr{M}$ has been mitigated and whether and how a new arbitrary test data instance d_t has been affected to debiasing when processed by $\mathscr{M}$. To this purpose, we can either rely on a self-explainable (interpretable) model for implementing $\mathscr{C}$ or take advantage of some post-hoc explanation method. In this paper, for the sake of generality, we propose adopting post-hoc XAI techniques, such as LIME or SHAP, to inspect model's behaviour and predictions. Assuming that SHAP is chosen as explanation

Table 1. Overview of the datasets used in the study.

Name	Sensitive	Target	Size ($n \times d$)
Adult [6]	sex, race	income	48,842 $\times$ 14
Bank Marketing [19]	marital	y	45,211 $\times$ 16
Credit Card [26]	marital	Y	30,000 $\times$ 23

method, we can compute a feature importance value for each $X_j \in X$ so that we can get an idea of which features are more important globally for $\mathscr{C}$ and locally for predictions $\hat{y}^{\mathscr{C}}$. Using SHAP, it is also possible to determine which values of the features contribute more to triggering the debiasing process.

4 Experiments & Discussion

In this section, we present and discuss the results of the experiments that we designed to assess our framework. The goal of our experiments is to show that our method can be used to obtain useful knowledge about the debiasing process under different experimental conditions. Before presenting the results in detail, we first describe the experimental settings, including the datasets, classification models, and explanation methods used.

4.1 Datasets

In the study, we used three datasets that are commonly employed in the Fair AI literature: **Adult** [6], which contains census data to classify whether the annual income of an individual is above 5OK$/year or not; **Bank Marketing** [19], which covers data related to marketing campaigns via phone calls of a Portuguese bank institution and whether the client subscribed or not to a term deposit; **Credit Card** [26] consisting of data that concern customers' default payments. The statistics of the datasets are presented in Table 1.

We preprocessed the data using a feature-wise pipeline. Missing values in numerical features are imputed using the mean, while categorical ones with the most frequent value. Numerical features are subsequently scaled using Min-Max normalisation, whereas categorical attributes are encoded through one-hot encoding (OHE).

4.2 Experimental Setting

Once a dataset has been preprocessed, it was split into two equally-sized parts: one part was used for training the mitigated classification model, and the other was used for probing it. This is because the mitigated model should not be tested with previously encountered instances. The training step of the classification model depends on the type of debiasing performed.

Given the three categories of mitigation techniques (pre-processing, in-processing and post-processing), we selected one algorithm for each type: *Prototype Representation Learner*[5] [27] for pre-processing, *Exponentiated Gradient* [6] [1] for in-processing, and *Threshold Optimizer* [7][14] as post-processing mechanism. Mitigation was performed for each specified sensitive attribute in the dataset and models' hyper-parameters were mapped to

[5]https://fairlearn.org/main/api_reference/generated/fairlearn.preprocessing.PrototypeRepresentationLearner.html
[6]https://fairlearn.org/v0.10/api_reference/generated/fairlearn.reductions.ExponentiatedGradient.html
[7]https://fairlearn.org/v0.10/api_reference/generated/fairlearn.postprocessing.ThresholdOptimizer.html

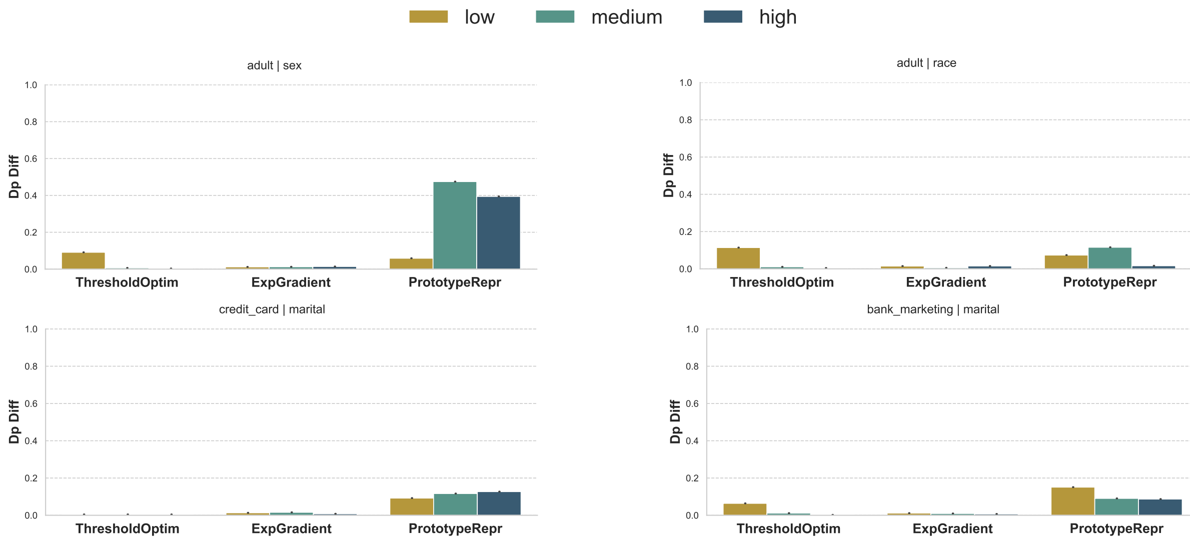


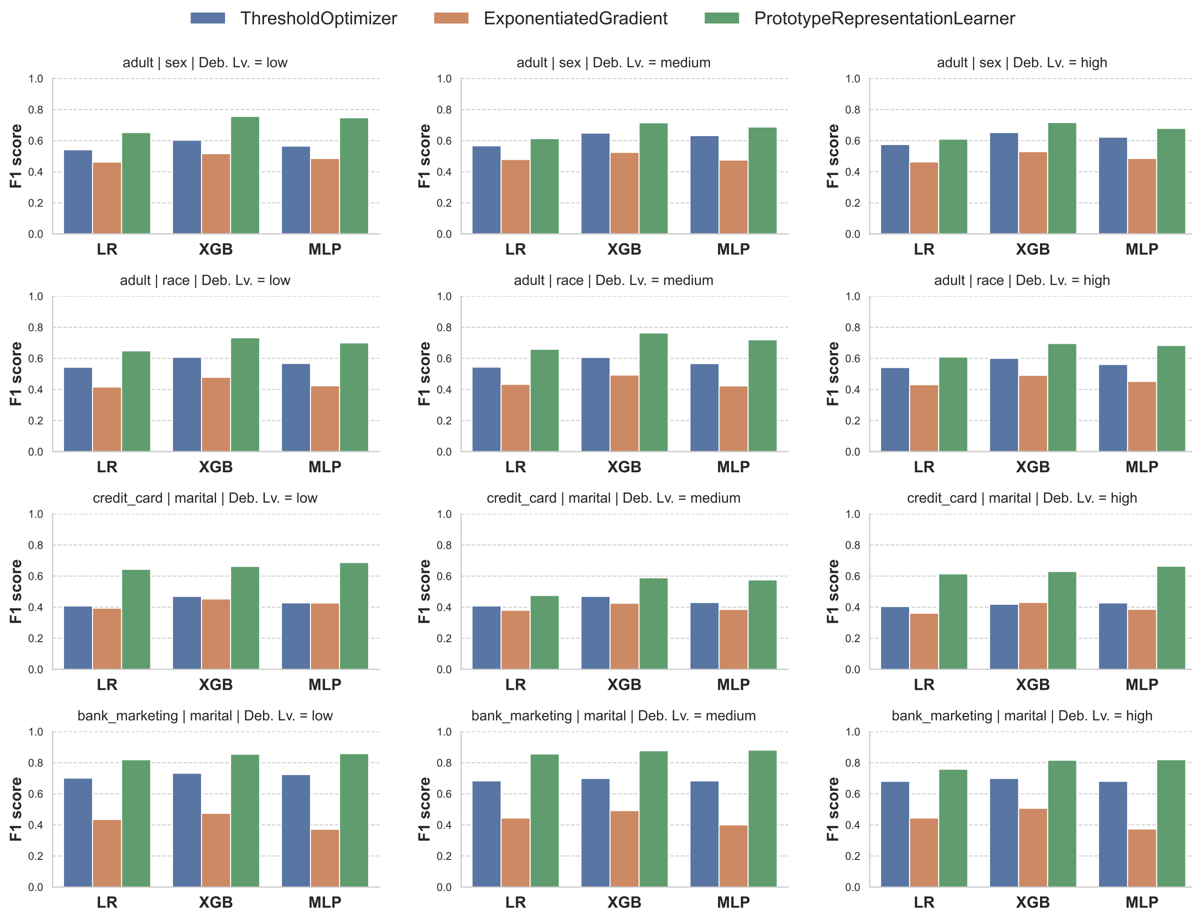
Fig. 3. **Fairness of the Mitigated model** w.r.t. different levels of fairness. On the top row, we report the Demographic Parity difference for *Adult - Sex* (left) and *Adult - Race* (right). On the bottom row, we show *Credit Card - Marital* (left) and *Bank Marketing - Marital* (right).

different levels of fairness (*Low*, *Medium*, and *High*). All implementations employed for this step come from the *Fairlearn* package [25].

For the pre-processing technique, various representations were created, and then learning upon them is performed using *Logistic Regression* models[8]. When using the in-process method, the mitigated model was learnt directly on the training/test sets determined by the original split. In the case of the post-processing approach, an unmitigated model was first learnt, and then passed to the mitigation procedure. After we have obtained the mitigated model, we treated it as a black-box and probed it with the aim of obtaining its predictions.

We then applied the label generation technique described in Equation 1 and obtained a new dataset, composed by the probing instances and the new determined labels. Additionally, since the resulting dataset was unbalanced, we applied an oversampling technique [17][9] to balance it and boost the performance of the classifier. Using this dataset, we trained three different discriminator models: *Logistic Regression*, *Multi-Layer Perceptron*[10] and *XGBoost*[11]. Each time a model was trained, a grid search was performed to tune its parameters and support model selection. Finally, we applied the Kernel SHAP explainer to the classifier. The background was computed using 100 samples, every feature of the dataset was taken into account, and 1000 simulations were used to estimate SHAP values.

[8]https://scikit-learn/stable/modules/generated/sklearn.linear_model.LogisticRegression.html
[9]https://imbalanced-learn.org/stable/references/generated/imblearn.over_sampling.RandomOverSampler.html
[10]https://scikit-learn/stable/modules/generated/sklearn.neural_network.MLPClassifier.html
[11]https://xgboost.readthedocs.io/en/release_3.2.0/



(a) Low (b) Medium (c) High

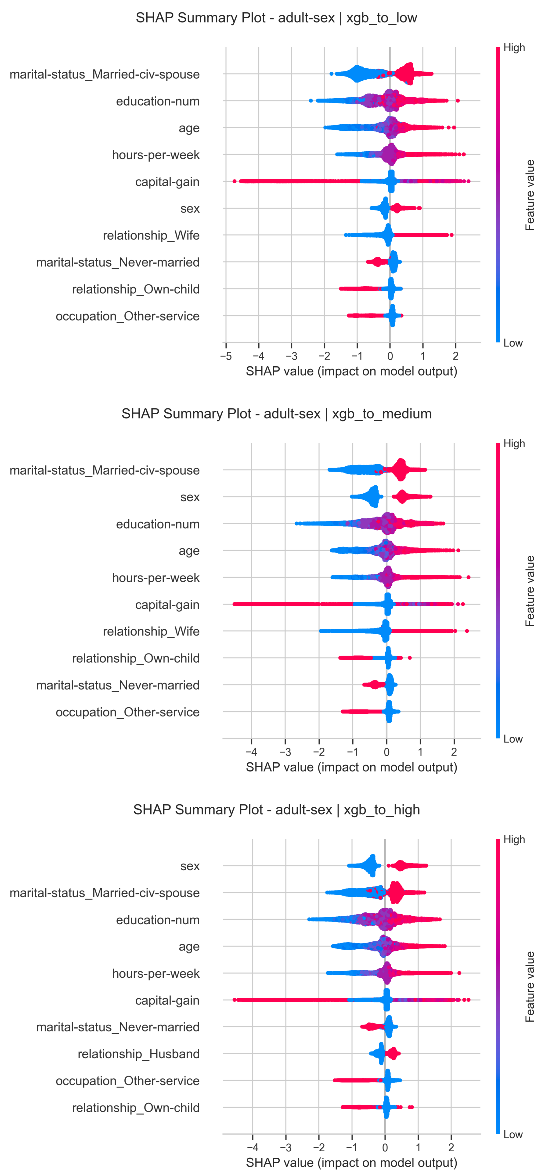
Fig. 4. **Accuracy of the Discriminator model** w.r.t. different levels of fairness. We report the F1 score for *Adult* - *Sex* in the first row, for *Adult* - *Race* in the second one, for *Credit Card* - *Marital* in the third and for *Bank Marketing* - *Marital* in the last one.

All experiments were executed in Python on a server with 32 Intel Xeon Skylake cores running at 2.1 GHz, 256 GB RAM. The source code for reproducing all our experiments is available online[12].

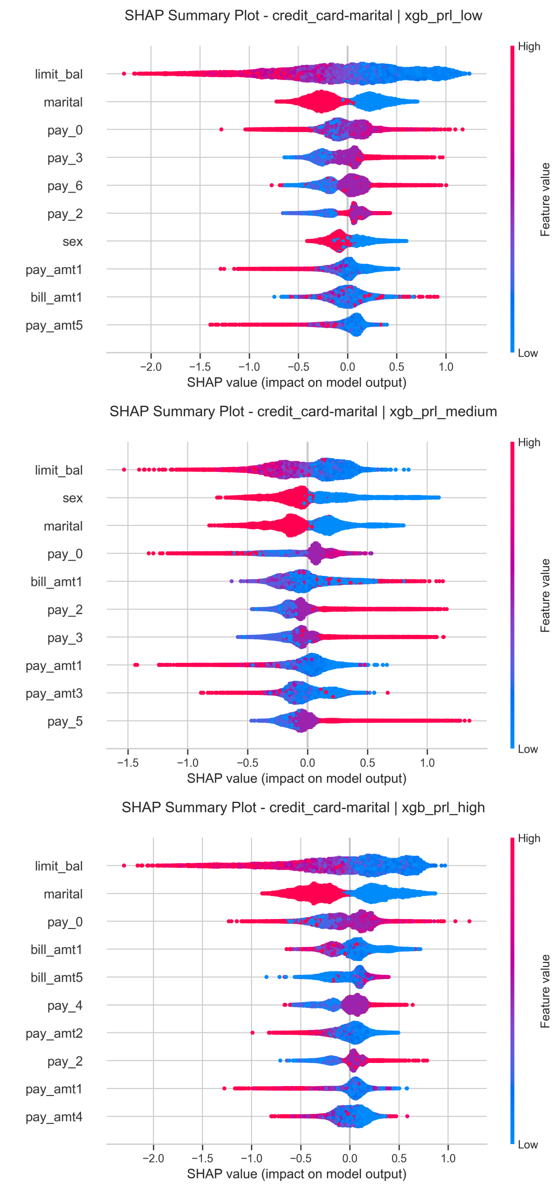
4.3 Results

First, we assess the behaviour of the mitigated model for all levels of fairness and with different sensitive features (Figure 3). We expect an increase in fairness and a loss in accuracy as the fairness level increases. To verify this, we

[12]https://github.com/alessiofamiani/deb-in-class



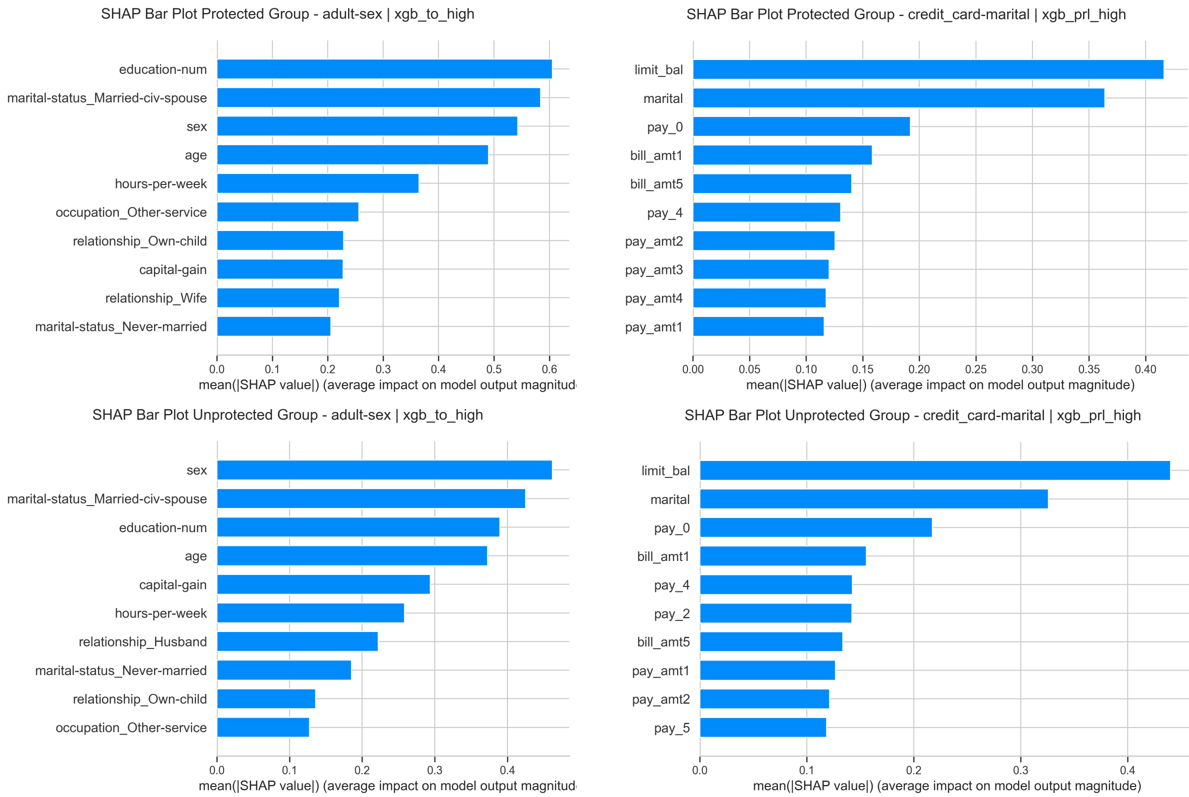
(a) Adult - Sex (TO)



(b) Credit Card - Marital (PRL)

Fig. 5. **SHAP applied on the discriminator** (XGBoost). The first row indicates a *Low* level of debiasing, middle row *Medium*, and last one *High*. SHAP summary plots are limited to the top 10 most important features.

computed the Demographic Parity [1, 2] to quantify the fairness (or unfairness), and the F1 score (for the positive label) to measure the accuracy of each model.



(a) Adult - Sex (TO)

(b) Credit Card - Marital (PRL)

Fig. 6. Feature importance for the protected (top) and unprotected (bottom) groups for XGBoost and with the highest level of debiasing.

As can be seen in Figure 3, *Exponentiated Gradient* and *Threshold Optimizer* induce a greater level of debiasing from the start, and perform generally better than *Prototype Representation Learner*. More specifically, *Exponentiated Gradient* seems to be the most effective and consistent method across all datasets, although *Threshold Optimizer* also appears to be consistent. In general, there is a decrease in accuracy as the fairness level increases, with the exception of *Prototype Representation Learner*, which in some datasets (see *Adult - Sex*) has spikes in Demographic Parity.

We then assess the behaviour of the discriminator, whose accuracy results can be seen in Figure 4. According to the plots, the models dealing with representations generated by *Prototype Representation Learner* perform better. This might be due to the fact that the data preserve more utility, as suggested by Figure 3, where we have more unfairness for *Prototype Representation Learner* than we have for other debiasing methods. Noticeably, the performances are similar across the different fairness levels. Models dealing with *Exponentiated Gradient* perform poorly w.r.t. other debiasing methods. In general, we observe that *XGBoost* and *MLP* gave more accurate results than *Logistic Regression*. For brevity, in the remainder, we refer only to the results obtained through *XGBoost*.

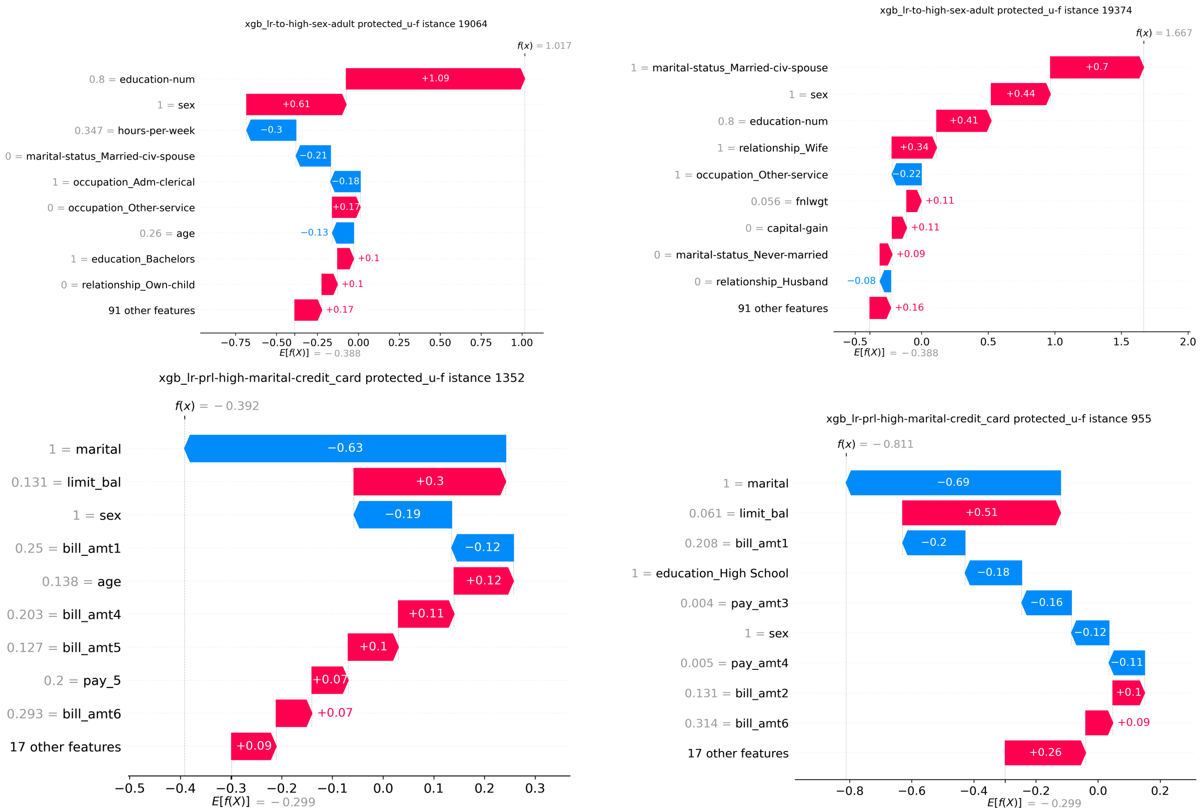


Fig. 7. SHAP plots for protected instances that goes from an unfavourable to a favourable outcome for XGBoost and with the highest level of debiasing. On the top row, instances belong to *Adult - Sex*, while on the bottom row they are related to *Credit Card - Marital*.

As final and most relevant analysis, we assess the explanation of the discriminator model. Some qualitative results about its behaviour are reported in Figure 5. In the *Adult* dataset mitigated on *sex*, we can observe that, when the debiasing level increases, the protected attribute gains more and more importance. For higher values of *sex*, indicating *Female*, chances for predicting the class *debiased* are greater. In addition, proxies and other features that could be sensitive are also ranked higher. Similarly, while looking at explanations for the *Credit card* dataset mitigated on *marital*, it can be seen that the protected attribute and its proxies are ranked higher, suggesting married males are increasing the chance of activating the debiasing.

We also plot feature importance for the protected groups and compare it with the one computed for the unprotected group (Figure 6). We can see that top features and their importance differ between demographic groups. This can be seen clearly in *Adult - Sex*, where women are impacted by features such as *relationship-Wife* and males by *relationship-Husband*. These differences support, as we expected, the idea that debiasing has a different impact on each group.

To go more in-depth, we also show the SHAP waterfall plots for random protected instances whose prediction flipped from an unfavourable outcome to a favourable one (Figure 7). In *Adult - Sex* the protected attributes – namely, the level of education and the marital status – greatly increase the chances of predicting the debiased class.

4.4 Limitations

Despite the encouraging results, it is important to acknowledge several limitations that may affect the feasibility and effectiveness of our approach. First, the method relies on the assumption that one has access to information about the input data (not necessarily those used to train the model), the ground-truth labels of the probing dataset, and the protected attributes. In circumstances where this assumption does not hold, the feasibility and utility of the method might be compromised. Even under ideal conditions, a significant limitation arises from the mild or poor performance of discriminator, primarily due to the unbalanced dataset used for training. In addition, tests are conducted on a limited number of datasets and the experimental workflow (e.g., the necessity of splitting the dataset into two parts) limits the suitability of other datasets generally employed in the Fair AI literature. This could be overcome by using synthetically generated data for the procedure. Moreover, while SHAP values provide useful insights for feature contributions, they remain correlational and do not imply causality. As a result, interpretations derived from these explanations should be treated with caution. Finally, feature importance of sensitive attributes can be suppressed as previous studies suggested [5, 7, 10].

5 Conclusions

In this work, we have analysed the impact of debiasing in the model's predictions and behaviour. We have proposed an original method to make the debiasing process less opaque by exploiting an arbitrary dataset to probe a black-box model, obtain its predictions, and try to capture the perturbation induced by the debiasing. By using SHAP on a classifier trained upon a new dataset collecting corrected (and not corrected) instances, we have analysed feature importance and drawn useful conclusions about the effectiveness and limitations of our framework.

Future work will address the known issues and drawbacks of our method. Additionally, we will extend the approach for supporting different data types other than tabular (i.e. images and text). We will also investigate methods to generate synthetic data, so that we could have more control in the analysis for testing and validating our approach. The label generation procedure could also be more refined, and other explanation techniques could be involved, including the adoption of an intrinsically interpretable discrimination model. Ultimately, it would be useful to try to quantify the overall impact the debiasing has on the predictions of a model, e.g., by defining a measure to assess the trade-off between fairness and accuracy.

References

[1] Alekh Agarwal, Alina Beygelzimer, Miroslav Dudík, John Langford, and Hanna M. Wallach. 2018. A Reductions Approach to Fair Classification. In *Proceedings of the 35th International Conference on Machine Learning, ICML 2018, Stockholmsmässan, Stockholm, Sweden, July 10-15, 2018*. PMLR, 60–69.

[2] Alekh Agarwal, Miroslav Dudík, and Zhiwei Steven Wu. 2019. Fair Regression: Quantitative Definitions and Reduction-Based Algorithms. In *Proceedings of the 36th International Conference on Machine Learning, ICML 2019, 9-15 June 2019, Long Beach, California, USA*. PMLR, 120–129.

[3] Ulrich Aïvodji, Hiromi Arai, Olivier Fortineau, Sébastien Gambs, Satoshi Hara, and Alain Tapp. 2019. Fairwashing: the risk of rationalization. In *Proceedings of the 36th International Conference on Machine Learning, ICML 2019, 9-15 June 2019, Long Beach,*

California, USA. PMLR, 161–170.

[4] Aparna Balagopalan, Haoran Zhang, Kimia Hamidieh, Thomas Hartvigsen, Frank Rudzicz, and Marzyeh Ghassemi. 2022. The Road to Explainability is Paved with Bias: Measuring the Fairness of Explanations. In *FAccT '22: 2022 ACM Conference on Fairness, Accountability, and Transparency, Seoul, Republic of Korea, June 21 - 24, 2022*. ACM, 1194–1206. doi:10.1145/3531146.3533179

[5] Hubert Baniecki and Przemyslaw Biecek. 2024. Adversarial attacks and defenses in explainable artificial intelligence: A survey. *Inf. Fusion* 107 (2024), 102303. doi:10.1016/J.INFFUS.2024.102303

[6] Barry Becker and Ronny Kohavi. 1996. Adult. UCI Machine Learning Repository. DOI: https://doi.org/10.24432/C5XW20.

[7] Tom Begley, Tobias Schwedes, Christopher Frye, and Ilya Feige. 2020. Explainability for fair machine learning. *CoRR* abs/2010.07389 (2020). https://arxiv.org/abs/2010.07389

[8] Jason A. Colquitt. 2001. On the dimensionality of organizational justice: A construct validation of a measure. *Journal of Applied Psychology* 86, 3 (2001), 386–400. doi:10.1037/0021-9010.86.3.386

[9] Jessica Dai, Sohini Upadhyay, Ulrich Aïvodji, Stephen H. Bach, and Himabindu Lakkaraju. 2022. Fairness via Explanation Quality: Evaluating Disparities in the Quality of Post hoc Explanations. In *AIES '22: AAAI/ACM Conference on AI, Ethics, and Society, Oxford, United Kingdom, May 19 - 21, 2021*. ACM, 203–214. doi:10.1145/3514094.3534159

[10] Botty Dimanov, Umang Bhatt, Mateja Jamnik, and Adrian Weller. 2020. You Shouldn't Trust Me: Learning Models Which Conceal Unfairness From Multiple Explanation Methods. In *Proceedings of the Workshop on Artificial Intelligence Safety, co-located with 34th AAAI Conference on Artificial Intelligence, SafeAI@AAAI 2020, New York City, NY, USA, February 7, 2020*. CEUR-WS.org, 63–73.

[11] Avijit Ghosh, Aalok Shanbhag, and Christo Wilson. 2022. FairCanary: Rapid Continuous Explainable Fairness. In *AIES '22: AAAI/ACM Conference on AI, Ethics, and Society, Oxford, United Kingdom, May 19 - 21, 2021*. ACM, 307–316. doi:10.1145/3514094.3534157

[12] Sofie Goethals, David Martens, and Toon Calders. 2024. PreCoF: counterfactual explanations for fairness. *Mach. Learn.* 113, 5 (2024), 3111–3142. doi:10.1007/S10994-023-06319-8

[13] Wei Guo, Yan Lyu, Xueyong Xu, and Weiwei Wu. 2024. Examining and Explaining Individual Fairness in Dynamic Pricing. In *40th International Conference on Data Engineering, ICDE 2024 - Workshops, Utrecht, Netherlands, May 13-16, 2024*. IEEE, 182–190. doi:10.1109/ICDEW61823.2024.00030

[14] Moritz Hardt, Eric Price, and Nati Srebro. 2016. Equality of Opportunity in Supervised Learning. In *Advances in Neural Information Processing Systems 29: Annual Conference on Neural Information Processing Systems 2016, December 5-10, 2016, Barcelona, Spain*. 3315–3323.

[15] Scott M. Lundberg and Su-In Lee. 2017. A Unified Approach to Interpreting Model Predictions. In *Advances in Neural Information Processing Systems 30: Annual Conference on Neural Information Processing Systems 2017, December 4-9, 2017, Long Beach, CA, USA*. 4765–4774.

[16] Ninareh Mehrabi, Umang Gupta, Fred Morstatter, Greg Ver Steeg, and Aram Galstyan. 2022. Attributing Fair Decisions with Attention Interventions. In *Proceedings of the 2nd Workshop on Trustworthy Natural Language Processing (TrustNLP 2022)*. ACL, Seattle, U.S.A., 12–25. doi:10.18653/v1/2022.trustnlp-1.2

[17] Giovanna Menardi and Nicola Torelli. 2014. Training and assessing classification rules with imbalanced data. *Data Min. Knowl. Discov.* 28, 1 (2014), 92–122. doi:10.1007/S10618-012-0295-5

[18] Christoph Molnar. 2022. *Interpretable machine learning: a guide for making black box models explainable* (second edition ed.). Christoph Molnar, Munich, Germany.

[19] S. Moro, P. Rita, , and P. Cortez. 2014. Bank Marketing. UCI Machine Learning Repository. DOI: https://doi.org/10.24432/C5K306.

[20] Romila Pradhan, Jiongli Zhu, Boris Glavic, and Babak Salimi. 2022. Interpretable Data-Based Explanations for Fairness Debugging. In *SIGMOD '22: International Conference on Management of Data, Philadelphia, PA, USA, June 12 - 17, 2022*. ACM, 247–261. doi:10.1145/3514221.3517886

[21] Marco Túlio Ribeiro, Sameer Singh, and Carlos Guestrin. 2016. "Why Should I Trust You?": Explaining the Predictions of Any Classifier. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, San Francisco, CA, USA, August 13-17, 2016*. ACM, 1135–1144. doi:10.1145/2939672.2939778

[22] Lloyd S Shapley. 1953. A Value for n-Person Games. In *Contributions to the Theory of Games II*. Princeton University Press, 307–317.

[23] Shubham Sharma, Jette Henderson, and Joydeep Ghosh. 2020. CERTIFAI: A Common Framework to Provide Explanations and Analyse the Fairness and Robustness of Black-box Models. In *AIES '20: AAAI/ACM Conference on AI, Ethics, and Society, New York, NY, USA,*

February 7-8, 2020. ACM, 166–172. doi:10.1145/3375627.3375812

[24] Christopher Starke, Janine Baleis, Birte Keller, and Frank Marcinkowski. 2022. Fairness perceptions of algorithmic decision-making: A systematic review of the empirical literature. *Big Data Soc.* 9, 2 (2022), 205395172211151. doi:10.1177/20539517221115189

[25] Hilde J. P. Weerts, Miroslav Dudík, Richard Edgar, Adrin Jalali, Roman Lutz, and Michael Madaio. 2023. Fairlearn: Assessing and Improving Fairness of AI Systems. *J. Mach. Learn. Res.* 24 (2023), 257:1–257:8.

[26] I-Cheng Yeh. 2009. Default of Credit Card Clients. UCI Machine Learning Repository. DOI: https://doi.org/10.24432/C55S3H.

[27] Richard S. Zemel, Yu Wu, Kevin Swersky, Toniann Pitassi, and Cynthia Dwork. 2013. Learning Fair Representations. In *Proceedings of the 30th International Conference on Machine Learning, ICML 2013, Atlanta, GA, USA, 16-21 June 2013*. JMLR.org, 325–333.

[28] Yuying Zhao, Yu Wang, and Tyler Derr. 2023. Fairness and Explainability: Bridging the Gap towards Fair Model Explanations. In *Thirty-Seventh AAAI Conference on Artificial Intelligence, AAAI 2023, Washington, DC, USA, February 7-14, 2023*. AAAI Press, 11363–11371. doi:10.1609/AAAI.V37I9.26344